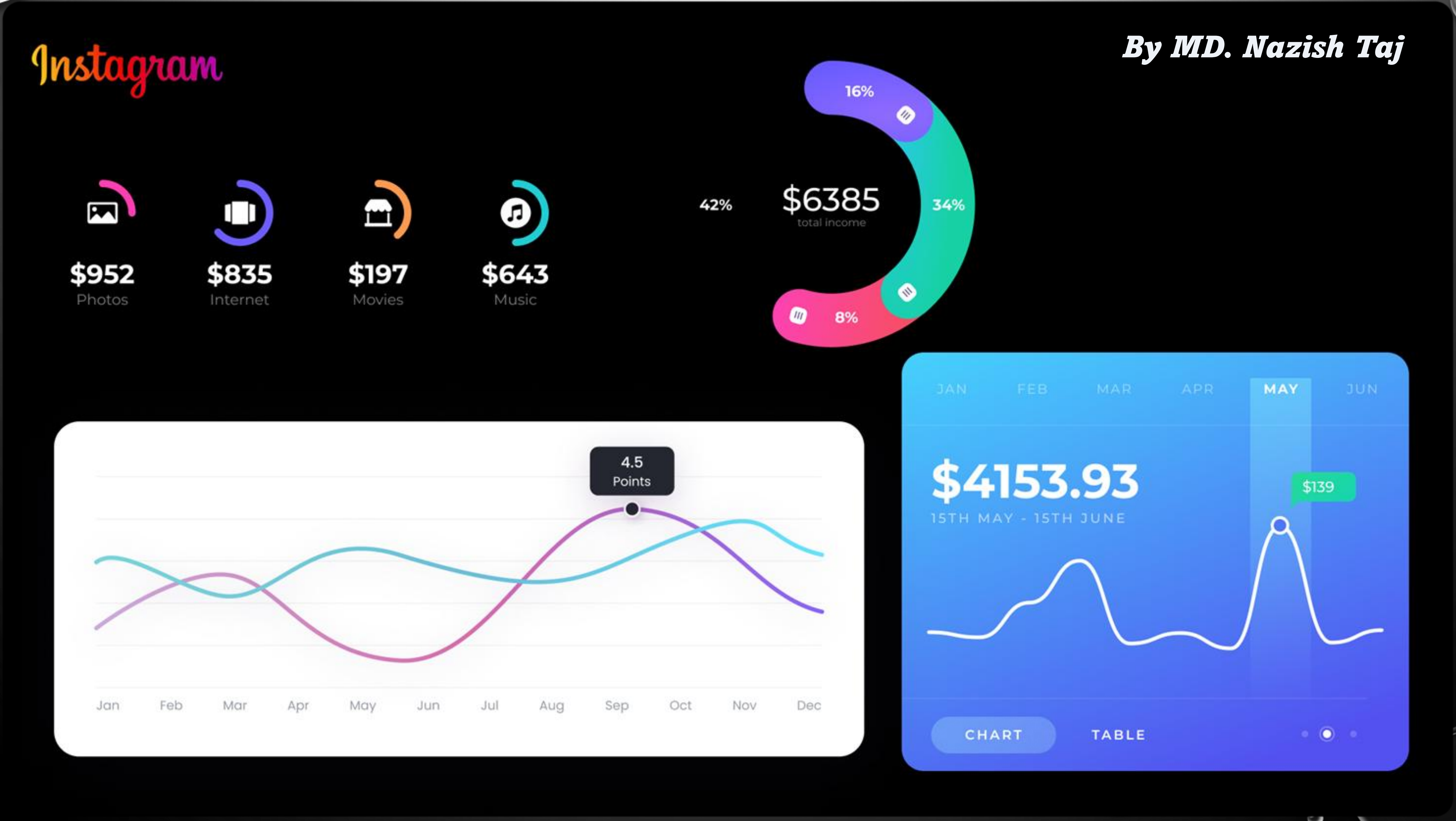
INSTAGRAM USER ANALYTICS

By MD. Nazish Taj



PROJECT DESCRIPTION: This project aims to analyze the raw data that is provided and retrieve useful insights from it by using SQL queries based on questions asked by the management team. Various database management tools can be used for this project.

PROJECT APPROACH: Using the provided RAW data to create a database in MYSQL Workbench and implementing various sorting and other queries for data extraction and valuable insights.

TECH-STACK USED: MySQL workbench, SQL queries , Microsoft PowerPoint

INSIGHTS:

A) **Marketing:**

1. Rewarding the Most Loyal Users: To find 5 oldest users of Instagram from the database provided

```
select * from users
order by created_at
limit 5;
```

select * **from** users
order by created_at
limit 5;

Result Grid | Filter Rows: | Edit:

id	username	created_at
80	Darby_Herzog	2016-05-06 00:14:21
67	Emilio_Bernier52	2016-05-06 13:04:30
63	Elenor88	2016-05-08 01:30:41
95	Nicole71	2016-05-09 17:30:22
38	Jordyn.Jacobson2	2016-05-14 07:56:26
NULL	NULL	NULL

Conclusion: Users with id 80 ,67,63,95,38 are 5 oldest users of Instagram

2. **Remind Inactive Users to Start Posting**: To find users who have never posted a single photo on Instagram.

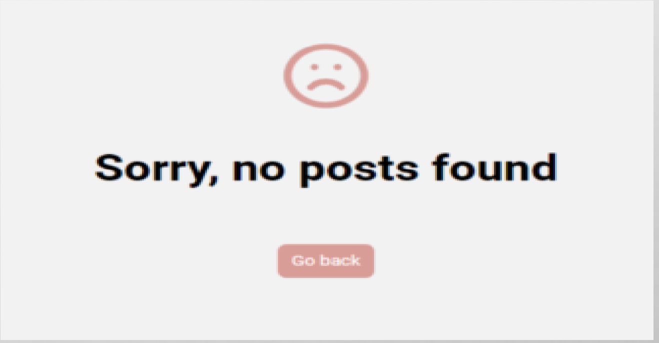
```
select users.username , photos.id from users
left join photos
on users.id=photos.user_id
where photos.id is Null;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

username	id
Aniya_Hackett	NULL
Kasandra_Homenick	NULL
Jaclyn81	NULL
Rocio33	NULL
Maxwell.Halvorson	NULL
Tierra.Trantow	NULL
Pearl7	NULL
Ollie_Ledner37	NULL
Mckenna17	NULL
David.Osinski47	NULL
Morgan.Kassulke	NULL
Linnea59	NULL
Duane60	NULL
Julien_Schmidt	NULL
Mike.Auer39	NULL
Franco_Keebler64	NULL
Nia_Haag	NULL
Hulda.Macejkovic	NULL

Result 117

select users.username , photos.id
from users
left join photos
on users.id=photos.user_id
where photos.id **is Null**;



Conclusion: Above mentioned users haven’t posted a single photo on Instagram.

3. Declaring Contest Winner: To find which user got most likes on a single photo

select users.username ,
likes.photo_id ,
count(likes.user_id) as Total_Likes
from likes
inner join photos **on**
photos.id=likes.photo_id
inner join users
on users.id = photos.user_id
group by likes.photo_id
order by Total_Likes desc
limit 1;

```
select  users.username , likes.photo_id , count(likes.user_id) as Total_Likes
from likes
inner join photos on
photos.id=likes.photo_id
inner join users on
users.id = photos.user_id
group by likes.photo_id
order by Total_Likes desc
limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

username	photo_id	Total_Likes
Zack_Kemmer93	145	48

Conclusion: Photo_ID 145 is the photo got with most likes posted by username Zack_Kemmer93.

4. **Hashtag Researching**: To identify and suggest Top 5 most commonly used tags on the platform.

select tags.tag_name ,
count(photo_id) as CountOfTags
from tags
inner join photo_tags
on tags.id=photo_tags.tag_id
group by photo_tags.tag_id
order by CountOfTags desc
limit 5;

```
select tags.tag_name , count(photo_id) as CountOfTags
from tags
inner join photo_tags
on tags.id=photo_tags.tag_id
group by photo_tags.tag_id
order by CountOfTags desc
limit 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content

tag_name	CountOfTags
beach	42
concert	24
fun	38
party	39
smile	59

Conclusion : smile , beach , party , fun and concert are the most used hashtags on the platform.

5**. Launch AD campaign**: To find which day most user registers on the platform.

select dayname(created_at)
as Day,
count(*) as No_Of_User_Registered
from users
group by dayname(created_at)
order by No_Of_User_Registereddesc;

```
select dayname(created_at) as Day, count(*) as No_Of_User_Registered
from users
group by dayname(created_at)
order by No_Of_User_Registered
desc;
```

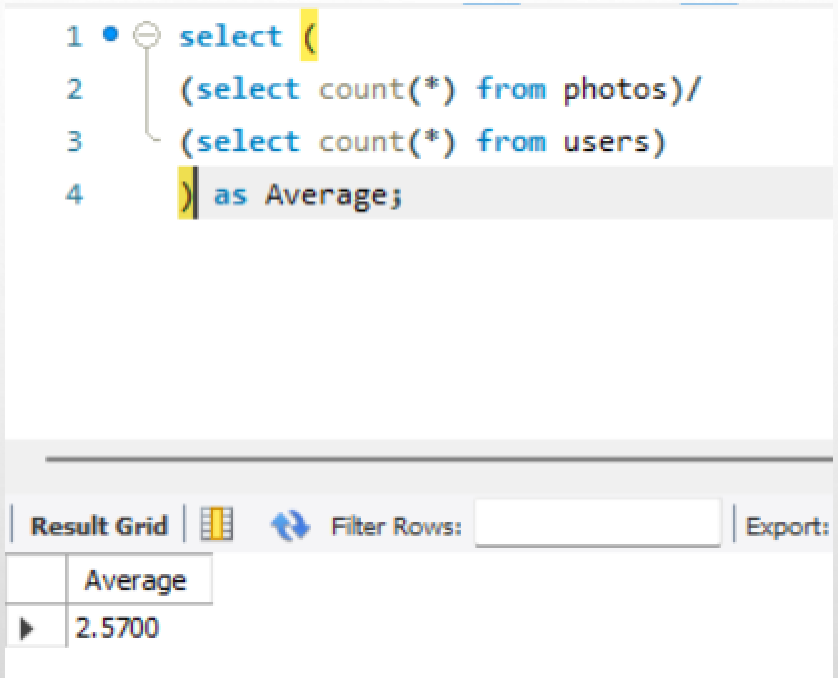
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Day	No_Of_User_Registered
Thursday	16
Sunday	16
Friday	15
Tuesday	14
Monday	14
Wednesday	13
Saturday	12

Conclusion: Most user registers on **Thursday and Sunday**.

B)INVESTOR METRICS

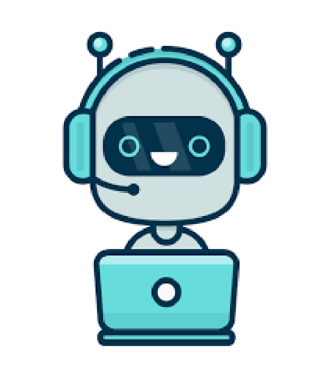
1. **User Engagement:** To find how many time on avg user post photos and also to show total no of photos and total no . of users.



select (
(select count(*) from photos)/
(select count(*) from users)
) as Average;

Conclusion : Average photos uploaded by user is 2.5700

2. **Bots & Fake account:** Identify users (potential bots) who have liked every single photo on the site, as this is not typically possible for a normal user..



select users.username,
count(photo_id) as TotalLikes
from likes
inner join users
on users.id=likes.user_id
group by user_id
having TotalLikes=(select count(*) from photos) **order** by TotalLikes desc;

```sql
select users.username, count(photo_id) as TotalLikes from likes
inner join users
on users.id=likes.user_id
group by user_id
having TotalLikes=(select count(*) from photos)
order by TotalLikes desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

username	TotalLikes
Aniya_Hackett	257
Jaclyn81	257
Rocio33	257
Maxwell.Halvorson	257
Ollie_Ledner37	257
Mckenna17	257
Duane60	257
Julien_Schmidt	257
Mike.Auer39	257
Nia_Haag	257
Leslie67	257
Janelle.Nikolaus81	257
Bethany20	257

Conclusion: Above mentioned users liked every single photos.

THANK YOU

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